

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Chapter 2: Probability

LEARNING OBJECTIVES

1. Sample Spaces and Events
2. Interpretations of Probability
3. Addition Rules
4. Conditional Probability
5. Multiplication and Total Probability Rules
6. Independence
7. Bayes' Theorem

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Definition

Random experiment

- An experiment that can result in different outcomes, even though it is repeated in the same manner every time, is called a **random experiment**.
- The set of all possible outcomes of a random experiment is called the **sample space** of the experiment. The sample space is denoted as S .
- An **event** is a subset of the sample space of a random experiment.

Sample spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Example

Random experiment: Roll a die

Sample space: $S = \{1, 2, 3, 4, 5, 6\}$

Event: $E_1 = \{\text{Die is even}\} = \{2, 4, 6\}$

$E_2 = \{\text{Die is odd}\} = \{1, 3, 5\}$



Sample spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Tree Diagrams

Sample spaces can also be described graphically with **tree diagrams**.

- When a sample space can be constructed in several steps or stages, we can represent each of the n_1 ways of completing the first step as a branch of a tree.
- Each of the ways of completing the second step can be represented as n_2 branches starting from the ends of the original branches, and so forth.

Sample spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

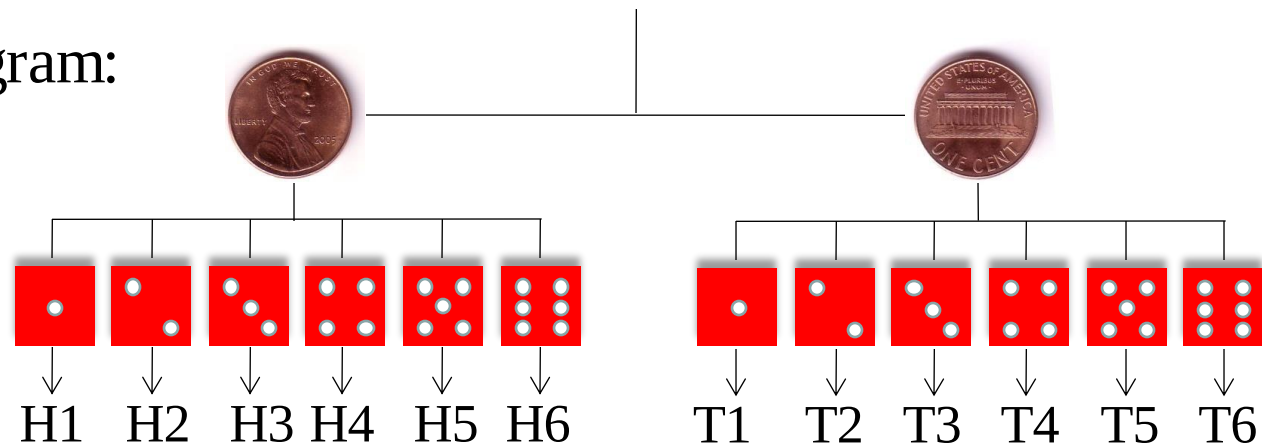
Bayes' Theorem

Summary

Example

A probability experiment consists of tossing a coin and then rolling a six-sided die. Describe the sample space.

Tree diagram:



The sample space has 12 outcomes:

$$S = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$$

Sample spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

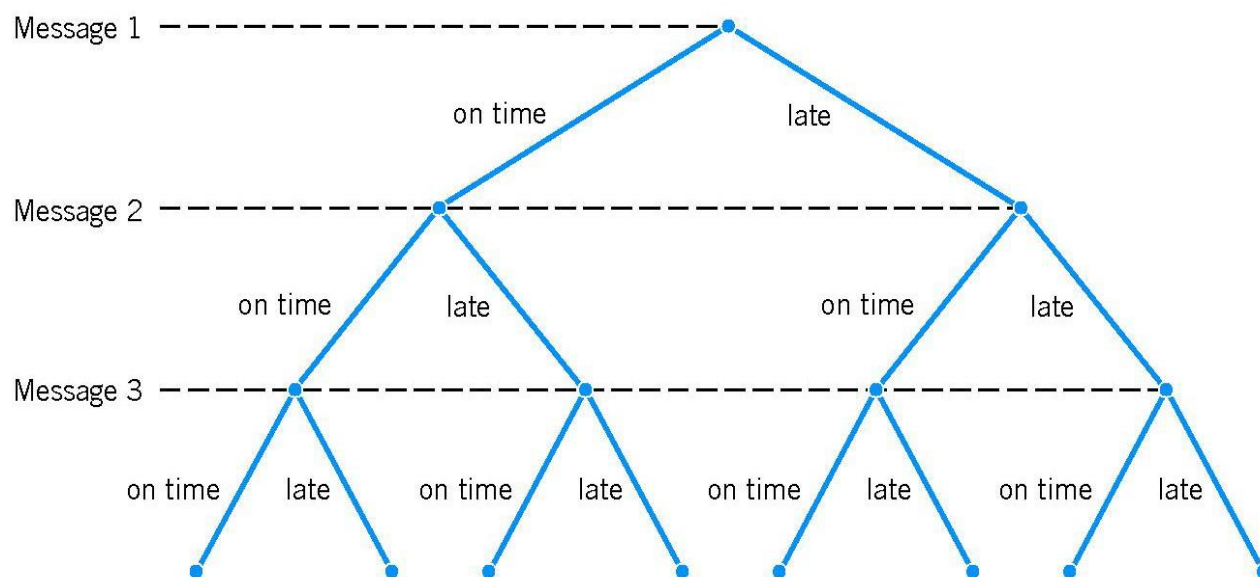
Independence

Bayes' Theorem

Summary

Example

Each message in a digital communication system is classified as to whether it is received within the time specified by the system design. If three messages are classified, use a tree diagram to represent the sample space of possible outcomes.



Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Basic Set Operations

The **union** of two events is the event that consists of all outcomes that are contained in either of the two events. We denote the union as $E_1 \cup E_2$.

The **intersection** of two events is the event that consists of all outcomes that are contained in both of the two events. We denote the intersection as $E_1 \cap E_2$.

The **complement** of an event in a sample space is the set of outcomes in the sample space that are not in the event. We denote the complement of the event E as E' .

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

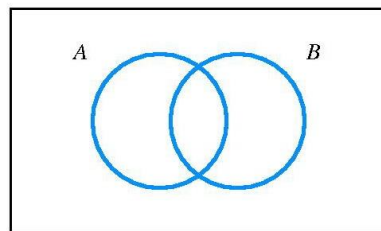
Independence

Bayes' Theorem

Summary

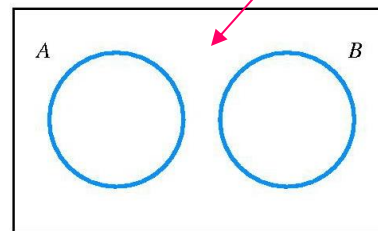
Venn Diagrams

mutually exclusive

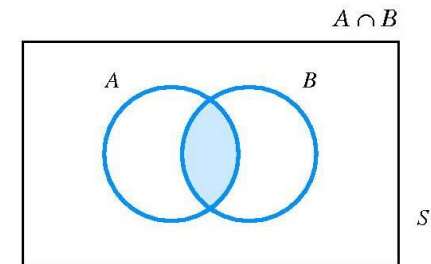


(a)

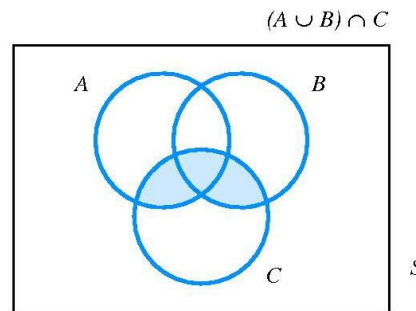
Sample space S with events A and B



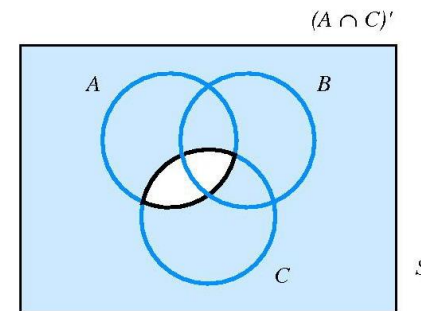
(b)



(c)



(d)



(e)

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Important properties:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

$$A = (A \cap B) \cup (A \cap B')$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Introduction

There are three approaches to assessing the probability of an uncertain event:

1. **a priori classical probability**: the probability of an event is based on prior knowledge of the process involved.
2. **empirical classical probability**: the probability of an event is based on observed data.
3. **subjective probability**: the probability of an event is determined by an individual, based on that person's past experience, personal opinion, and/or analysis of a particular situation.

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Equally Likely Outcomes

Whenever a sample space consists of N possible outcomes that are equally likely, the probability of each outcome is $1/N$.



1. **a priori classical probability**

$$\text{Probability of Occurrence} = \frac{X}{T} = \frac{\text{number of ways the event can occur}}{\text{total number of possible outcomes}}$$

2. **empirical classical probability**

$$\text{Probability of Occurrence} = \frac{\text{number of favorable outcomes observed}}{\text{total number of outcomes observed}}$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Example priori classical probability

Find the probability of selecting a face card (Jack, Queen, or King) from a standard deck of 52 cards.

$$\text{Probability of Face Card} = \frac{X}{T} = \frac{\text{number of face cards}}{\text{total number of cards}}$$

$$\frac{X}{T} = \frac{12 \text{ face cards}}{52 \text{ total cards}} = \frac{3}{13}$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Example empirical classical probability

Find the probability of selecting a male taking statistics from the population described in the following table:

	Taking Stats	Not Taking Stats	Total
Male	84	145	229
Female	76	134	210
Total	160	279	439

$$\begin{aligned}
 \text{Probability of Male Taking Stats} &= \frac{\text{number of males taking stats}}{\text{total number of people}} \\
 &= \frac{84}{439} = 0.191
 \end{aligned}$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Axioms of Probability

Probability is a number that is assigned to each member of a collection of events from a random experiment that satisfies the following properties:

If S is the sample space and E is any event in a random experiment,

$$(1) P(S) = 1$$

$$(2) 0 \leq P(E) \leq 1$$

(3) For two events E_1 and E_2 with $E_1 \cap E_2 = \emptyset$

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

The special addition rule

1. If A and B are mutually exclusive events,

$$P(A \cup B) = P(A) + P(B)$$

2. A collection of events, $E_1, E_2, \dots, E_k$ is said to be **mutually exclusive** if for all pairs,

$$E_i \cap E_j = \emptyset$$

For a collection of mutually exclusive events,

$$P(E_1 \cup E_2 \cup \dots \cup E_k) = P(E_1) + P(E_2) + \dots + P(E_k)$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

The general addition rule

1. Two events: A and B are any events,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

2. Three events:

$$\begin{aligned} P(A \cup B \cup C) = & P(A) + P(B) + P(C) - P(A \cap B) \\ & - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C) \end{aligned}$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Example

Find the probability of selecting a male or a statistics student from the population described in the following table:

	Taking Stats	Not Taking Stats	Total
Male	84	145	229
Female	76	134	210
Total	160	279	439

$$\begin{aligned}
 P(\text{Male or Stat}) &= P(M) + P(S) - P(M \text{ or } S) \\
 &= 229/439 + 160/439 - 84/439 = 305/439
 \end{aligned}$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Conditional Probability

To introduce conditional probability, consider an example involving manufactured parts.

Let D denote the event that a part is defective and let F denote the event that a part has a surface flaw.

Then, we denote the probability of D given, or assuming, that a part has a surface flaw as $P(D|F)$. This notation is read as the **conditional probability** of D given F , and it is interpreted as the probability that a part is defective, given that the part has a surface flaw.

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Example 2-22(page 42)

Table 2-3 Parts Classified

		Surface Flaws		
		Yes (event F)	No	Total
Defective	Yes (event D)	10	18	38
	No	30	342	362
	Total	40	360	400

$$P(D|F) = 10/40 = 0.25$$

$$P(D|F') = 18/360 = 0.05$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

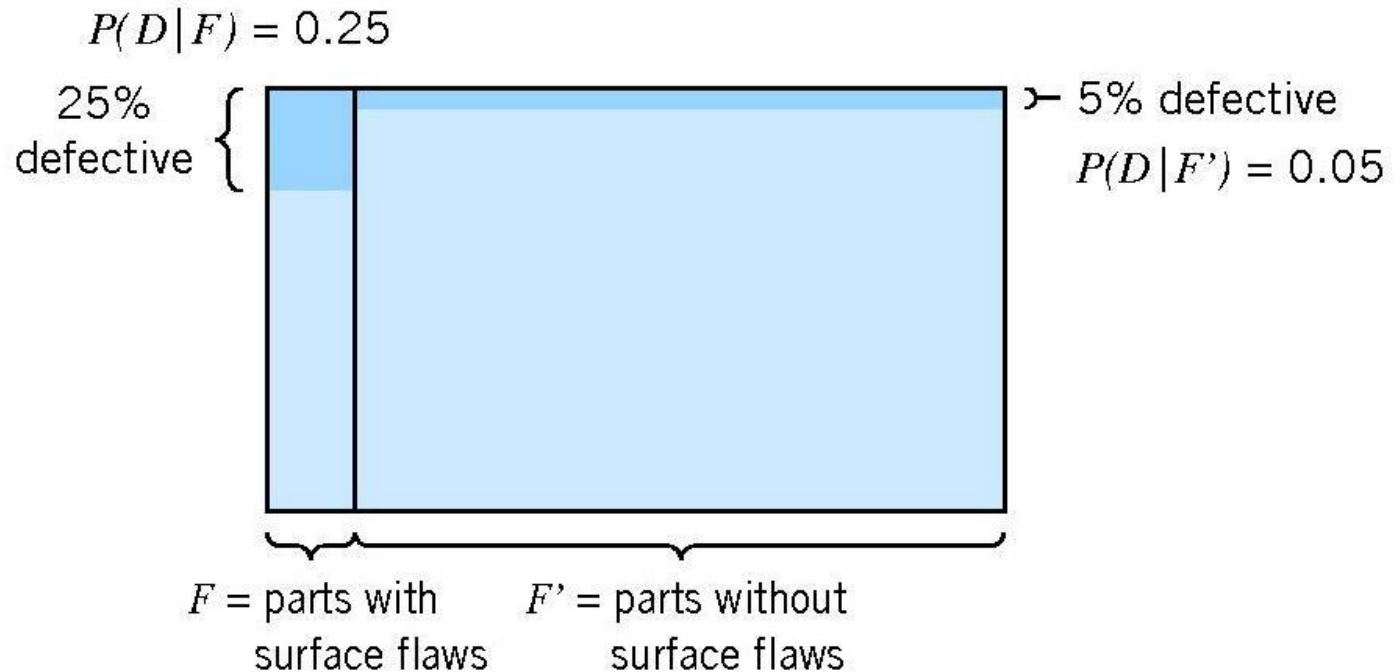
Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary



Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Definition

Conditional Probability

The conditional probability of an event B given an event A , denoted as $P(B|A)$, is

$$P(B|A) = P(B \cap A)/P(A)$$

for $P(A) > 0$.

Special case: all outcomes are equally likely

$$P(A \cap B)/P(A) = \frac{\text{number of outcomes in } A \cap B}{\text{number of outcomes in } A}$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Example 2-23 (page 43)

$$P(D|F) = P(D \cap F)/P(F) = \frac{10}{400} / \frac{40}{400} = \frac{10}{40}$$

Note that in this example all four of the following probabilities are different:

$$P(F) = 40/400 \quad P(F|D) = 10/28$$

$$P(D) = 28/400 \quad P(D|F) = 10/40$$

Here, $P(D)$ and $P(D|F)$ are probabilities of the same event, but they are computed under two different states of knowledge. Similarly, $P(F)$ and $P(F|D)$ are computed under two different states of knowledge.

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Example

Of the cars on a used car lot, 70% have air conditioning (AC) and 40% have a CD player (CD). 20% of the cars have both.

What is the probability that a car has a CD player, given that it has AC ?

	CD	No CD	Total
AC	0.2	0.5	0.7
No AC	0.2	0.1	0.3
Total	0.4	0.6	1.0

$$\begin{aligned}
 P(\text{CD} \mid \text{AC}) &= \frac{P(\text{CD and AC})}{P(\text{AC})} \\
 &= \frac{0.2}{0.7} = .2857
 \end{aligned}$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Multiplication Rule

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

Example

The probability that an automobile battery subject to high engine compartment temperature suffers low charging current is 0.7. The probability that a battery is subject to high engine compartment temperature is 0.05.

The probability that a battery is subject to low charging current and high engine compartment temperature is

$$P(C \cap T) = P(C|T)P(T) = 0.7 \times 0.05 = 0.035$$

$C = \{\text{a battery suffers low charging current}\}$

$T = \{\text{a battery is subject to high engine compartment temperature}\}$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Partition of an event

Figure 2-15

Partitioning an event into two mutually exclusive subsets.

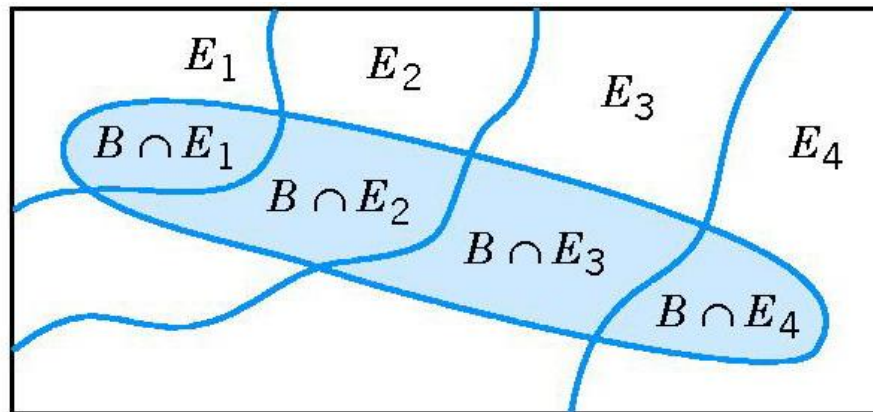
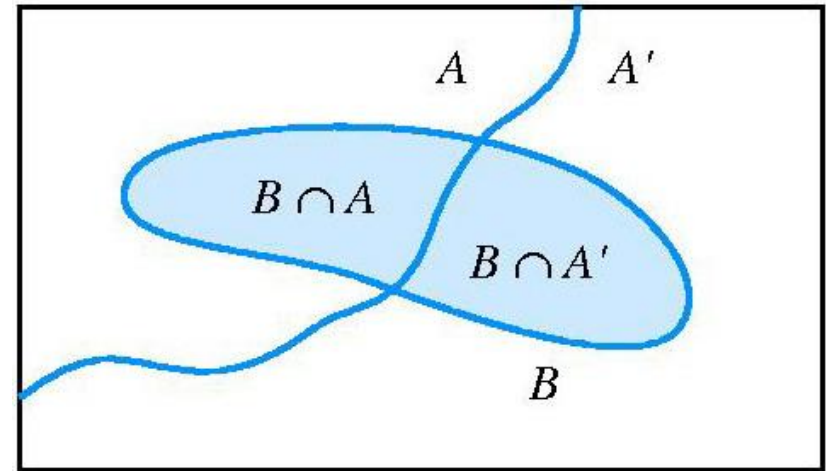


Figure 2-16

Partitioning an event into several mutually exclusive subsets.

$$B = (B \cap E_1) \cup (B \cap E_2) \cup (B \cap E_3) \cup (B \cap E_4)$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Total Probability Rule: two events

$$P(B) = P(BA) + P(BA') = P(B|A)P(A) + P(B|A')P(A')$$

Total Probability Rule: multiple events

$$E_1 \cup E_2 \cup \dots E_k = S$$

Assume $E_1, E_2, \dots, E_k$ are k mutually exclusive and exhaustive sets. Then

$$\begin{aligned} P(B) &= P(B \cap E_1) + P(B \cap E_2) + \dots + P(B \cap E_k) \\ &= P(B|E_1)P(E_1) + P(B|E_2)P(E_2) + \dots + P(B|E_k)P(E_k) \end{aligned}$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Example 2-27 (page 48)

Consider the contamination discussion at the start of this section. The information is summarized here

Probability of Failure	Level of Contamination	Probability of Level
0.1	High	0.2
0.005	Not High	0.8

Let F denote the event that the product fails, and let H denote the event that the chip is exposed to high levels of contamination. The requested probability is $P(F)$, and the information provided can be represented as

$$\begin{aligned}
 P(F|H) &= 0.10 & \text{and} & & P(F|H') &= 0.005 \\
 P(H) &= 0.20 & \text{and} & & P(H') &= 0.80
 \end{aligned}$$

From Equation 2-11,

$$P(F) = 0.10(0.20) + 0.005(0.80) = 0.024$$

which can be interpreted as just the weighted average of the two probabilities of failure.

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Definition

Two events is called independent if any one of the following equivalent statements is true:

$$(1) P(A|B) = P(A)$$

$$(2) P(A \cap B) = P(A)P(B)$$

$$(3) P(B|A) = P(B)$$

Proposition: If A and B are independent events, then so are events A and B' , events A' and B , and events A' and B' .

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Example 2-31 (page 52)

A day's production of 850 manufactured parts contains 50 parts that do not meet customer requirements. Two parts are selected at random, without replacement, from the batch. Let $A = \{\text{the first part is defective}\}$, and let $B = \{\text{the second part is defective}\}$.

We suspect that these two events are not independent because knowledge that the first part is defective suggests that it is less likely that the second part selected is defective.

$$P(B|A) = 49/849$$

$$\begin{aligned} P(B) &= P(B|A)P(A) + P(B|A')P(A') \\ &= (49/849)(50/850) + (50/849)(800/850) \\ &= 50/850 \end{aligned}$$

the two events
are not
independent, as
we suspected.

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Definition

The events $E_1, E_2, \dots, E_n$ are independent if and only if for any subset of these events $E_{i_1}, E_{i_2}, \dots, E_{i_k}$

$$P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_k}) = P(E_{i_1})P(E_{i_2}) \dots P(E_{i_k})$$

Exercise

Two coins are tossed. Let A denote the event “at most one head on the two tosses,” and let B denote the event “one head and one tail in both tosses.” Are A and B independent events?

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

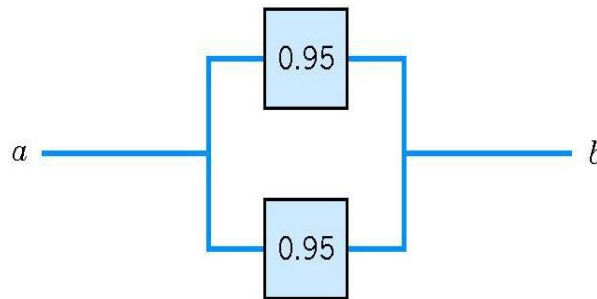
Independence

Bayes' Theorem

Summary

Example 2-34 (page 53)

The following circuit operates only if there is a path of functional devices from left to right. The probability that each device functions is shown on the graph. Assume that devices fail independently. What is the probability that the circuit operates?



$$\begin{aligned} P(T \cup B) &= 1 - P[(T \cup B)'] \\ &= 1 - P(T' \cap B') \end{aligned}$$

$$\begin{aligned} P(T' \cap B') &= P(T')P(B') \\ &= (1-0.95)^2 = 0.05^2 \end{aligned}$$

$$P(T \cup B) = 1 - 0.05^2 = 0.9975$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Bayes' Theorem

If $E_1, E_2, \dots, E_k$ are k mutually exclusive and exhaustive events and B is any event,

$$P(E_1 | B) = \frac{P(B | E_1)P(E_1)}{P(B | E_1)P(E_1) + P(B | E_2)P(E_2) + \dots + P(B | E_k)P(E_k)} \quad \text{for } P(B) > 0$$

In special case:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} \quad \text{for } P(B) > 0$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Example

In a state where cars have to be tested for the emission of pollutants, 25% of all cars emit excessive amount of pollutants. When tested, 99% of all cars that emit excessive amount of pollutants will fail, but 17% of all cars that do not emit excessive amount of pollutants will also fail. What is the probability that a car that fails the test actually emits excessive amounts of pollutants?

Let A denote the event that a car fails the test and B the event that it emits excessive amounts of pollutants:

$$P(B) = 0.25, P(A|B) = 0.99 \text{ and } P(A|B') = 0.17.$$

We have to find $P(B|A)$?

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

$$P(B | A) = \frac{P(A | B)P(B)}{P(A)}$$

where

$$\begin{aligned} P(A) &= P(AB) + P(AB') = P(A|B)P(B) + P(A|B')P(B') \\ &= 0.375 \end{aligned}$$

Substitution into the formula for $P(B|A)$ yields

$$P(B|A) = 0.66$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Example

Two events A and B are such that $P[A \cap B] = 0.15$, $P[A \cup B] = 0.65$, and $P[A|B] = 0.5$. Find $P[B|A]$.

Solution $P[A \cup B] = P[A] + P[B] - P[A \cap B] \Rightarrow 0.65 = P[A] + P[B] - 0.15$. This means that $P[A] + P[B] = 0.65 + 0.15 = 0.80$. Also, $P[A \cap B] = P[B] \times P[A|B]$. This then means that

$$P[B] = \frac{P[A \cap B]}{P[A|B]} = \frac{0.15}{0.50} = 0.30$$

Thus, $P[A] = 0.80 - 0.30 = 0.50$. Since $P[A \cap B] = P[A] \times P[B|A]$, we have that

$$P[B|A] = \frac{P[A \cap B]}{P[A]} = \frac{0.15}{0.50} = 0.30$$

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Definition

A **random variable** is a function that assigns a real number to each outcome in the sample space of a random experiment.

A random variable is denoted by an uppercase letter such as X . After an experiment is conducted, the measured value of the random variable is denoted by a lowercase letter such as x 70 milliamperes.

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

Definition

A **discrete** random variable is a random variable with a finite (or countably infinite) range.

A **continuous** random variable is a random variable with an interval (either finite or infinite) of real numbers for its range.

Sample Spaces

Probability

Addition Rules

Conditional Pro.

Multi. Rules

Total Pro. Rule

Independence

Bayes' Theorem

Summary

We have studied:

1. Sample Spaces and Events
2. Interpretations of Probability
3. Addition Rules
4. Conditional Probability
5. Multiplication and Total Probability Rules
6. Independence
7. Bayes' Theorem

Homework: Read slides of the next lecture.